

From chopped to cooked: Design and inference in physical environments

Anonymous CogSci submission

Abstract

Physical spaces are often designed to support specific uses. But how do people create such environments, and how do users infer their intended function? We propose that design and inference about design are complementary processes, grounded in a capacity to mentally simulate goal-directed actions. We tested this using “Overcooked”-style kitchens where participants either judged what a kitchen was designed for (Study 1) or designed kitchens for cooks with varying goals and beliefs (Study 2). In Study 1, participants inferred that kitchens were designed for tasks the layout made easier to complete, consistent with the prediction of a simulation-based computational model. In Study 2, participants made designs that helped cooks efficiently complete their task, adjusting their choices when cooks faced uncertainty about which task to perform. Together, these findings point towards a study of design as a cognitive activity grounded in the same mechanisms that support planning and social reasoning.

Keywords: design; mental simulation; social cognition; Bayesian inference; inverse planning

Introduction

Imagine opening a new restaurant. How would you design the kitchen? If you need to prepare a general-purpose buffet, you might arrange the equipment to support a wide variety of dishes. But if you were opening an Italian restaurant, you might specialize certain areas for specific tasks—for example, positioning flour and water within easy reach to streamline pasta-making. The same logic applies beyond restaurants. Suppose you are an urban planner designing an airport. Anticipating that travelers may run late, you might design wide corridors with clear signage so that people do not run into each other or get lost.

Designing Environments

The study of environment design has a long history in architecture and urban planning, where it has given rise to many intuitions about how to make an environment “well designed”. For example, the notion that *form follows function*—that good design responds systematically to human needs—has received considerable attention in design theory and practice (Corbusier, 1923; D. Norman, 2013). In cognitive science, a smaller body of work has examined how people rearrange objects to reduce visual search times and facilitate foraging behaviors (Gray et al., 2006; Solman & Kingstone, 2017). While these studies demonstrate that people are sensitive to the relationship between spatial structure and task efficiency, so far no unifying computational framework has been

developed that makes quantitative predictions about how people structure environments to support their goals.

Inferring Intent from Design

If environments reflect intentional design, people who recognize this should be able to infer their intended use and act accordingly. Consider a chef who has just been hired at a new restaurant. Walking into a kitchen where pasta-making equipment is prominently positioned and easily accessible, they will likely infer the restaurant’s specialty. Similarly, a traveler navigating an unfamiliar airport can see that the wide corridors and prominent signage were placed deliberately, and trust these features to guide their path rather than wandering at random. Among design practitioners, this communicative dimension is called *legibility*: the idea that environments should make their function apparent to users (Montgomery, 1998; D. A. Norman, 1986).

In cognitive science, recent work has examined how people use Theory of Mind to extract social information from physical traces in the environment—for instance, inferring from objects in front of a doorway that others should not enter (Jara-Ettinger & Schachner, 2024; Lopez-Brau & Jara-Ettinger, 2023; Teo et al., 2025). Yet little work has examined how people reason about durably designed environments like kitchens and airports, which exist not just to *communicate* information about their use, but also to *support* users’ goals.

The Present Research

In this work, we consider design and inference about design as complementary processes, grounded in broader capacities for reasoning about agents’ beliefs, goals, and plans. We hypothesize that people design environments by reasoning about the goals and epistemic states of their users. We explore these ideas using kitchen environments inspired by the video game *Overcooked*, in which cooks navigate gridworld kitchens to prepare dishes (Figure 1).

In Study 1, we ask whether people can infer what a kitchen was designed for from its spatial arrangement. In Study 2, we investigate how people design kitchen layouts for particular purposes, taking into account the goals and beliefs of the user. We find that people’s inferences and design decisions are largely captured by a model grounded in planning, with some evidence that designers also consider users’ beliefs.

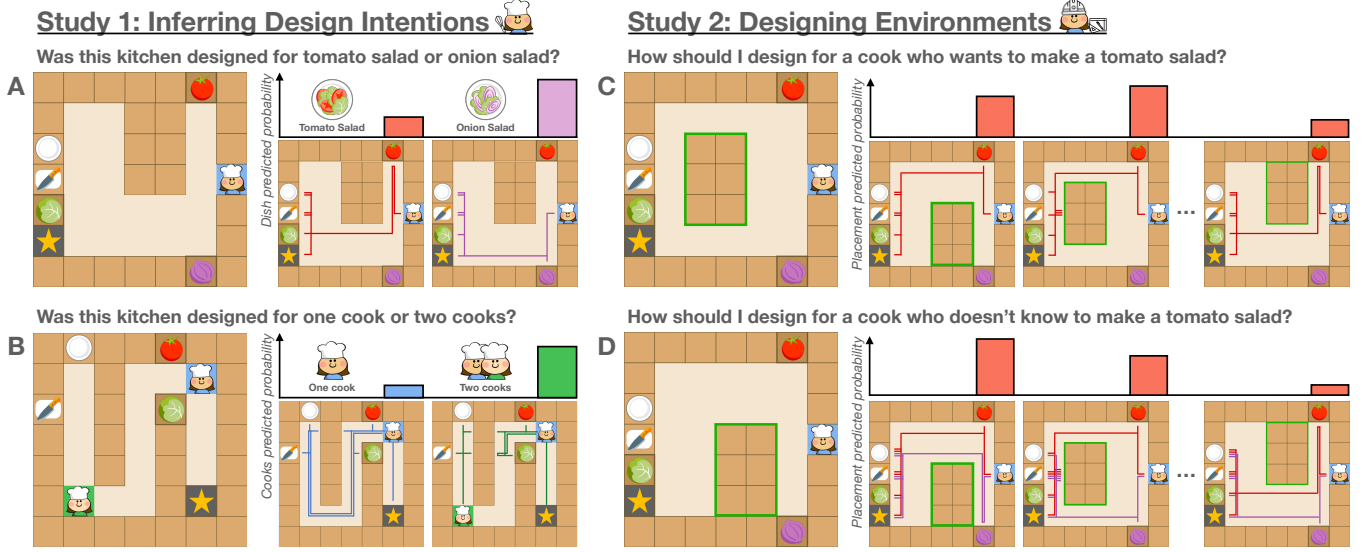


Figure 1: **Overview of Approach.** Each panel shows a kitchen with paths illustrating a cook’s planning; bar plots show illustrative model predictions. (A)–(B) *Inference task.* Participants inferred whether kitchens were designed for a particular dish or number of cooks; the model assigns higher probability to whichever option requires fewer steps. (C)–(D) *Design task.* Participants placed furniture to help cooks who knew or did not know which dish to make; possible furniture placements are shown. In (C), the model assigns higher probability to placements that help the cook complete the dish in fewer steps; in (D), the model additionally considers whether placements signal which dish the cook should make.

Study Environment

In the *Overcooked* environment, cooks attempt to complete one of two possible dishes in as few steps as possible. Each kitchen is a 7×7 grid containing several functional objects: *ingredient dispensers* that supply tomatoes, onions, and lettuce; *cutting boards* where ingredients can be chopped; a *plate*; and a *delivery window* where completed dishes are submitted. Cooks navigate the grid by moving in the four cardinal directions. They cannot move through counters or occupy the same space as another cook. They can carry one item at a time and must put items down on surfaces to free their hands.

To complete an order, cooks must gather the required ingredients, chop each one, combine them on a plate, and deliver the plated dish. We use two recipes that differ only in one ingredient: *tomato salad* (chopped tomato + chopped lettuce) and *onion salad* (chopped onion + chopped lettuce). Submitting the wrong dish resets the kitchen, forcing the cook to start over.

Computational Model

We develop a computational framework for modeling how people infer design intent and design environments for others.

Simulation model

We formalize the cook’s planning as a Markov Decision Process (MDP; Bellman, 1957). A kitchen configuration $\mathcal{M}$ defines an MDP in which the cook takes actions to complete a dish in as few steps as possible. Let $C_{\mathcal{M}}(g)$ denote the minimum number of steps required to complete goal g in con-

figuration $\mathcal{M}$. We estimate this using a hierarchical planner that decomposes recipes into subtasks and greedily selects the next subtask based on which has the lowest cost (see Wu et al., 2021, for full formalism).

When the kitchen has two cooks, we consider two models of multi-agent coordination. In the *independent* planner, each cook plans greedily, treating the other as a moving obstacle. In the *coordinating* planner, cooks use inverse planning to infer each other’s current subtask and avoid duplicating effort (see the ‘Bayesian Delegation model’ in Wu et al., 2021).

Modeling inference about design intent

An observer who encounters a designed environment can infer what it was designed for by considering which tasks it makes easier to complete. If a kitchen makes tomato salad easier to prepare than onion salad, an observer should infer it was designed for tomato salad. We model this inference as a softmax over negative step counts:

$$P_{\mathcal{M}}(g) \propto \exp(-\lambda \cdot C_{\mathcal{M}}(g)), \quad (1)$$

where g indexes possible goals (e.g., tomato vs. onion salad) and λ controls how strongly the inference tracks efficiency differences.

Modeling environment design

In the design task (Study 2), participants place a 2×3 furniture block at position k within a base layout ℓ , yielding 12 possible configurations; we denote the resulting configuration as $\mathcal{M}_{\ell,k}$. We model designers as selecting placements accord-

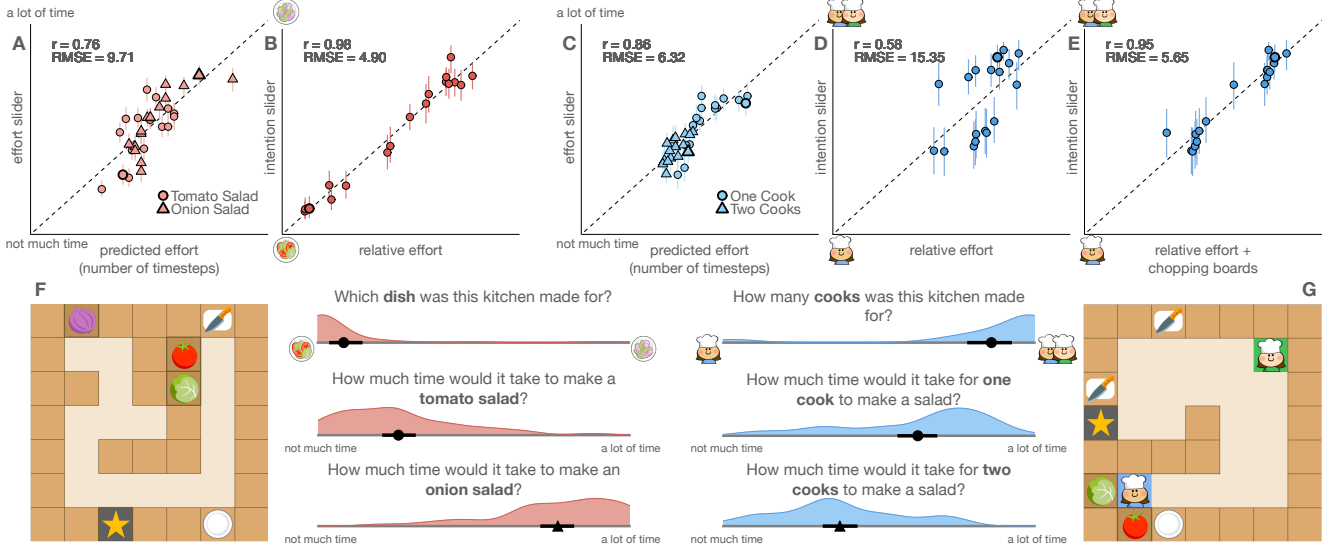


Figure 2: **Inference Study Results.** Each scatterplot shows one point per kitchen layout, comparing model predictions with mean participant responses; error bars show 95% CIs. Bolded points correspond to the example trials in (F)–(G). (A)–(B) ‘Dish condition’. Effort estimates correlated with the planner’s predicted step counts (A), and intention inferences tracked relative effort (B). (C)–(E) ‘Cooks condition’. Effort estimates correlated with model predictions (C), but intention inferences were only moderately predicted by relative effort alone (D); adding the number of cutting boards improved fit (E). (F)–(G) Example trials from the dish (F) and cooks (G) conditions, with response distributions shown.

ing to a softmax over utilities:

$$P(k | \ell, g) \propto \exp(\beta \cdot U_{\mathcal{M}_{\ell,k}}(g)), \quad (2)$$

where g is the intended goal and β controls how strongly participants prefer high-utility placements. When the cook knows which dish to make, the designer only needs to select placements that minimize step costs, giving the *Oracle* model: $U_{\mathcal{M}_{\ell,k}}(g) = -C_{\mathcal{M}_{\ell,k}}(g)$.

Trading off efficiency and communicativity. Consider a cook who does not know which dish to make and attempts the wrong one first. That cook incurs the full cost of completing the wrong dish, then must start over and complete the correct one. A designer who anticipates this possibility should evaluate each placement not only by how efficiently the intended dish can be completed, but also by how likely the cook is to attempt the wrong dish first. We formalize this with a utility function that captures both considerations:

$$U_{\mathcal{M}_{\ell,k}}(g) := -C_{\mathcal{M}_{\ell,k}}(g) - (1 - P_{\mathcal{M}_{\ell,k}}(g)) \cdot C_{\mathcal{M}_{\ell,k}}(\neg g), \quad (3)$$

where $\neg g$ denotes the alternative dish and $P_{\mathcal{M}_{\ell,k}}(g)$ is the probability the cook attempts the correct dish first (Equation 1). The first term captures the *efficiency* of completing the intended dish. The second term captures the *communicative cost*: the expected wasted effort if the cook misinfers and must restart. This is our *Full* model.

Alternative models. We compare the Full and Oracle models against two ablations. The *Inference-Ablation* model assumes

the cook guesses randomly, so $P_{\mathcal{M}_{\ell,k}}(g) = 0.5$. The *Efficiency-Ablation* model retains the cook’s efficiency-based inference but removes sensitivity to step costs: $U_{\mathcal{M}_{\ell,k}}(g) = P_{\mathcal{M}_{\ell,k}}(g)$.

Study 1: Inferring Design Intent

If environments are designed to support specific uses, observers should be able to infer what a space is for by noticing which tasks it makes easier. We tested this hypothesis by collecting two kinds of judgments. One group estimated how much effort different tasks would require in each kitchen; another judged what each kitchen was designed for. We predicted that intention inferences reflect assessments of relative task difficulty, as formalized in our computational model (Equation 1). We examined this in two contexts: inferring which dish a kitchen was designed for (tomato vs. onion salad) and how many cooks it was designed for (one vs. two)¹.

Methods

Participants Participants were recruited via Prolific. A total of 200 participants (*age*: median = 39, range = 20–88; *gender*: 102 female, 96 male, 2 non-binary; *race*: 17 Black/African American, 14 Asian, 1 American Indian/Alaska Native, 7 multiracial, 156 white, 5 other) completed the study. Participants were evenly assigned into one of four between-subjects conditions in a 2 (dish vs. cooks) $\times$ 2 (effort vs. inference) design. All participants were native

¹Materials, preregistrations, and data are available [here](#). Experiments were coded using jsPsych (De Leeuw et al., 2023). AI tools were used in the preparation of the materials.

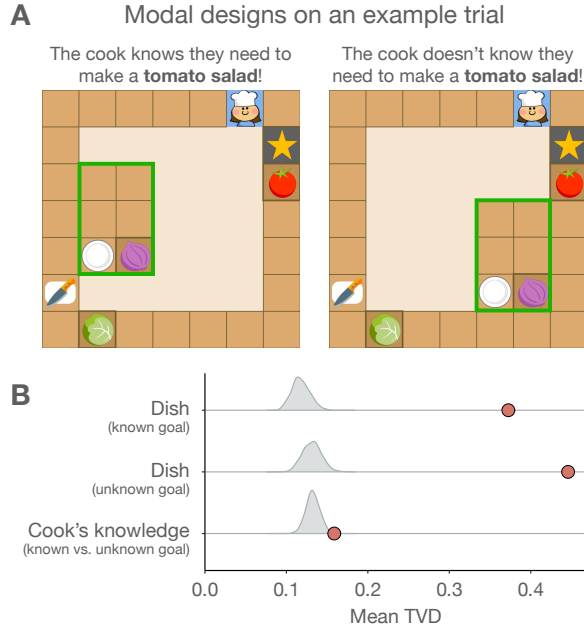


Figure 3: Design Study Results. (A) *Modal placements on an example trial.* In the known-goal condition, participants placed the furniture on the left-hand side. In the unknown-goal condition, participants blocked the onion, ensuring the cook could only make the intended dish. (B) Permutation tests comparing placement distributions (gray = null distribution; red = total variation distance (TVD) between placement distributions). Participants placed furniture differently depending on the target dish (top two rows) and, albeit much less so, depending on whether the cook knew which dish to make (bottom row).

English speakers residing in the US. Participants were paid \$2.40 for an estimated 12 minutes to complete the study in the effort conditions, and \$3.00 for an estimated 15 minutes to complete the study in the inference conditions (*mean completion time*: 13.0 mins).

Stimuli We created 18 kitchen layouts for each condition. In the ‘dish condition’, kitchens contained one tomato, one onion, and one lettuce dispenser, allowing preparation of either salad. In the ‘cooks condition’, kitchens contained only a tomato and lettuce dispenser, and showed starting positions for two possible cooks. Layouts were hand-crafted to span a range of design intentions, and varied in the number of interior countertops (4, 6, or 9) and cutting boards (1 or 2).

Procedure Participants first learned about the Overcooked environment through a series of instructions. Participants in the effort conditions were asked to “estimate how long it would take to make a dish in each kitchen”. Participants in the inference conditions were asked to “judge how each kitchen was designed to be used”. In the ‘dish condition’, participants were told that some kitchens were designed for a cook to make an *Onion Salad* while others were designed

for a *Tomato Salad*. In the ‘cooks condition’, they were told that some kitchens were designed for a cook to work alone while others were designed for two cooks to collaborate. Participants then completed a comprehension check containing questions about the environment mechanics; incorrect responses returned them to the instructions. All participants passed the comprehension check, though some required multiple attempts (*mean attempts*: 1.0, *range*: 0–7).

Participants viewed 18 kitchen layouts in randomized order. On each trial, they saw a static image of the kitchen with the cook’s starting position marked. Participants responded using sliders to rate either effort or design intent (Figure 2F–G shows example trials with response formats).

Results

For each kitchen layout, we compared the mean effort ratings and the mean intention ratings across participants. We asked whether participants’ effort estimates aligned with our simulation model, and whether their judgments about what each kitchen was designed for tracked how much easier one task seemed than the other

Inferring what dish a kitchen was designed for In the ‘dish condition’, participants judged whether each kitchen was designed for making a tomato salad or an onion salad (Figure 2F). We first compared participants’ effort estimates with our hierarchical planner, which approximates the minimum number of steps an agent would need to complete each dish. Participants’ mean effort ratings were correlated with simulated step counts for both tomato salad ($r = 0.68$, $p = .004$) and onion salad ($r = 0.82$, $p < .001$), suggesting that participants’ estimates of task difficulty aligned with those of a rational planner (Figure 2A).

We then asked whether these effort estimates could account for participants’ intention inferences. For each kitchen, we computed two quantities: the average rating of what it was designed for, and the average difference in perceived effort between dishes. We then fit a binomial regression predicting intention from relative effort. Intention inferences were strongly predicted by relative effort ($r = 0.98$, $p < .001$; Figure 2B): kitchens judged easier for making tomato salad were also judged as designed for it. When predicting participants’ intention inferences using the *planner’s* relative step counts, we find a similar pattern ($r = 0.80$, $p < .001$).

Inferring how many cooks a kitchen was designed for In the ‘cooks condition’, participants judged whether each kitchen was designed for one cook working alone or for two cooks collaborating to make a tomato salad (Figure 2G). For one cook, participants’ effort ratings were correlated with the number of steps the model required ($r = 0.78$, $p < .001$; Figure 2C). For two cooks, participants’ effort estimates were better captured by the coordinating planner ($r = 0.70$, $p = .003$) than the independent planner ($r = 0.65$, $p = .006$), so we used the former for subsequent analyses.

Participants’ judgments of how many cooks a kitchen was designed for were correlated with their effort estimates—

specifically, how much faster two cooks would be than one—but this correspondence was weaker than in the dish condition ($r = 0.58$, $p < .001$; Figure 2D). One possibility is that participants also drew on a simpler cue: the number of cutting boards in each kitchen. A kitchen with two cutting boards may signal that it was designed for two cooks, without requiring observers to mentally simulate how cooks would move through the space. Consistent with this possibility, intention ratings were well-predicted by a regression model combining both relative effort and the number of cutting boards as predictors ($r = 0.98$, $p < .001$; Figure 2E). This suggests that participants combined their assessments of task difficulty with this more direct cue when inferring how many cooks each kitchen was designed for.

When using the planner’s relative step counts to predict intention inferences, efficiency alone was similarly not a strong predictor of participants’ ratings ($r = 0.45$, $p < .001$). However, combining model predictions with the number of cutting boards yielded a strong fit ($r = 0.95$, $p < .001$), mirroring the pattern observed with participants’ own effort estimates.

Study 2: Designing Environments

Study 1 showed that observers can infer what a kitchen was designed for by assessing which tasks it makes easier. We now ask how people *design* environments for others to use. Unlike observers, who need only assess what an environment supports, designers must anticipate how users will act—a task that requires representing not just physical constraints but also users’ mental states. When users know their goal, design can focus on efficiency; when users are uncertain, the environment must also help them figure out *what* to do.

To test whether designers are sensitive to users’ epistemic states, we manipulated what participants believed about the cook who would use their kitchen. In the ‘known-goal’ condition, the cook would know which dish to make. In the ‘unknown-goal’ condition, the cook would not know which dish to make and would have to infer it from the environment. If designers consider users’ uncertainty, we should observe different patterns of design choices across conditions. Designers should select layouts that not only support efficient task completion but also signal which task to perform.

Methods

Participants Participants were recruited via Prolific. A total of 200 participants (*age*: median = 38, range = 18–69; *gender*: 112 female, 85 male, 2 non-binary, 1 other; *race*: 29 Black/African American, 23 Asian, 3 American Indian/Alaska Native, 1 Native Hawaiian/Pacific Islander, 5 multiracial, 139 white) completed the study, divided evenly across two between-subjects conditions: ‘known-goal’ and ‘unknown-goal’. All participants were native English speakers residing in the US and were paid \$3 for an estimated 15 minutes (*mean completion time*: 15.5 mins). Both studies reported here followed the Stanford University IRB protocol. All participants provided informed consent.

Table 1: **Design Study Results.** Δelpd : difference in expected log predictive density from leave-one-out cross-validation, relative to best model (with standard error; more negative = worse). Top-1 Acc. (%): proportion of trials where people select the model’s modal response. *Human*: split-half noise ceiling measuring human agreement. 95% CIs from bootstrap resampling over design problems.

Model	Δelpd (se)	Top-1 Acc. (%)
<i>Known-goal</i>		
Oracle	0 (0)	30.2 (23.6, 37.8)
Full	−98.4 (14.8)	23.9 (16.8, 30.9)
Inf. Abl.	−9.3 (5.3)	32.2 (27.0, 38.1)
Eff. Abl.	−1810.7 (34.2)	8.4 (5.1, 13.2)
Uniform	−1356.7 (19.4)	8.3 (8.3, 8.3)
<i>Human</i>	—	41.6 (39.8, 42.9)
<i>Unknown-goal</i>		
Oracle	0 (0)	25.2 (19.0, 32.6)
Full	−68.2 (14.9)	22.2 (16.5, 28.7)
Inf. Abl.	−40.3 (7.1)	25.9 (20.5, 32.0)
Eff. Abl.	−1540.5 (41.8)	7.7 (4.7, 11.2)
Uniform	−1097.9 (31.8)	8.3 (8.3, 8.3)
<i>Human</i>	—	40.1 (38.6, 41.0)

Stimuli We created 12 base kitchen layouts. Each layout had a 5×5 interior with fixed objects (delivery window, plate, ingredient dispensers, cutting board): some positioned around the perimeter, others attached to a movable furniture block. Participants chose where to place a 2×3 table within the interior; this table could occupy any of 12 valid positions. Each participant completed 24 design trials: all 12 layouts crossed with both target dishes (tomato salad and onion salad), presented in fully randomized order.

Procedure Participants first learned about the Overcooked environment through illustrated instructions. Critically, they were told that submitting the wrong dish causes the kitchen to reset, requiring the cook to start over. In the ‘known-goal’ condition, participants were told the cook would know which dish to make. In the ‘unknown-goal’ condition, participants were told the cook would not know which dish to make and would have to infer which one to try first. Participants then completed a comprehension check testing understanding of both the environment mechanics and the condition-specific manipulation (*mean attempts*: 0.4, *range*: 0–8).

Participants then completed three familiarization trials where they controlled a cook to prepare dishes. Trials ended upon successful delivery of the intended dish. In the ‘known-goal’ condition, participants saw the target recipe; in the ‘unknown-goal’ condition, they saw both recipes and guessed which to submit first. To ensure participants in the ‘unknown-goal’ condition understood the cost of submitting the wrong dish, we designed the final familiarization trial so that participants would *necessarily* submit the wrong dish first and experience restarting.

On each design trial, participants saw a base kitchen layout,

both possible dishes (with the target dish highlighted), and the furniture block. They dragged the furniture around the interior and could reposition it freely before submitting. In the ‘unknown-goal’ condition, each trial included a reminder that the cook does not know which dish to make.

Model fitting and evaluation. We fit each computational model separately for each experimental condition. For the cook’s inference (Equation 1), we fix $\lambda = 1$ to avoid double-fitting non-linear parameters in our models. We estimate a posterior distribution over the softmax sensitivity parameter β for each model using the R package *brms* (Bürkner, 2017). We then compute expected log predictive density (elpd) via approximate leave-one-out cross-validation, treating each placement decision as a separate observation.

Results

Designing for a cook who knows which dish to make In the ‘known-goal’ condition, participants’ furniture placements differed reliably depending on which dish the kitchen was designed for. We quantified this by computing the total variation distance (TVD) between placement distributions for tomato salad versus onion salad within each layout, testing significance via permutation of dish label within each kitchen (Figure 3B; $\text{TVD} = 0.373, p < .001$).

We then compared models using Δelpd and top-1 accuracy (see Table 1). Cross-validated model comparison strongly favored the Oracle model over a Uniform baseline ($\Delta\text{elpd} = -1356.69 \pm 19.38$), confirming that participants valued designs that made the intended dish more efficient to make. However, top-1 accuracy did not clearly discriminate among models: Oracle, Full, and Inference-Ablation all had overlapping confidence intervals. This suggests that while models differed in how they distributed probability across placements, they often agreed on which placement was most likely. A split-half noise ceiling shows that humans were more consistent with each other than with any model, but the models capture a substantial portion of the systematic variance.

Designing for a cook who does not know which dish to make In the ‘unknown-goal’ condition, participants’ furniture placements similarly differed depending on the target dish ($\text{TVD} = 0.446, p < .001$). Participants also made different design decisions across conditions, though this effect was smaller than the differences by dish (Figure 3B; $\text{TVD} = 0.159, p = .008$). This suggests that participants represented the cook’s uncertainty over which dish to make, and that this influenced their design choices. Figure 3A illustrates this pattern: in the ‘known-goal’ condition, participants placed furniture to minimize steps, whereas in the ‘unknown-goal’ condition, participants blocked access to the onion dispenser, ensuring the cook could *only* make the intended dish.

However, model comparisons did not reveal the predicted pattern (Table 1). We hypothesized the Full model would outperform the Oracle when cooks must infer their goal, but cross-validated comparison favored the Oracle in both conditions ($\Delta\text{elpd} = -68.16 \pm 14.92$). As in the known-goal condi-

tion, top-1 accuracy showed overlapping confidence intervals among the top models, suggesting that the models converged on similar modal predictions despite differing in their full distributions. The Efficiency-Ablation model performed substantially worse ($\Delta\text{elpd} = -1472.35 \pm 51.92$), indicating that efficiency remained central to participants’ design choices even when cooks faced uncertainty. Although participants reliably made different designs when cooks faced uncertainty, our models did not capture this difference.

Discussion

We asked whether design and inference about design are grounded in reasoning about agents taking efficient actions to achieve their goals. We found evidence for this in both directions. Designers created environments that helped users complete the intended task. When designers believed users were uncertain about their goal, they adjusted their designs accordingly. Observers inferred design intent by assessing which tasks an environment made relatively easier to complete. In both cases, a model that draws on simulation of goal-directed action captured key patterns in people’s judgments.

One limitation is that our design task did not clearly discriminate between models with an explicit communicative component and those reasoning only about efficiency. When a placement was highly optimized for one dish, it tended to be less suited for the other. As a result, efficiency-focused and communication-sensitive models often favored similar choices. This may also reflect our modeling approach: we implemented only shallow recursive reasoning. A deeper model might consider not just how well a placement supports the intended goal, but how distinctively it does so compared to other options the designer could have chosen instead (Gerstenberg & Tenenbaum, 2017; Goodman & Frank, 2016). Studying environments where efficiency and legibility trade off more directly could better discriminate between models.

Separately, our studies focused on environments designed to support efficient action. But design can serve other functions as well. A classroom, for example, might also be designed to prevent injury, reduce distraction, and scaffold learning. Designers might invest effort upfront to make an environment’s purpose self-evident, helping users more readily infer details about an environment’s use from its structure (Gibson, 2014; Kirsh, 1996; Rubio-Fernandez et al., 2025). Design practitioners capture this in the principle that “good design is unobtrusive” (Lovell, 2011). Extending the current framework to examine how people design environments that scaffold cognition is a promising direction.

Finally, our paradigm treated design as a one-shot choice among discrete options. Real-world design is often iterative, involves ill-defined objectives, and is shaped by shared conventions (Goldschmidt, 1991; Rittel & Webber, 1973; Simon, 2019). Understanding how people navigate larger design spaces through exploration and revision remains an important direction for building a more complete account of design as a cognitive activity.

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